

Sequences & Patterns



Arithmetic sequences

- 1 Find the 10th term of the sequence 5, 8, 11, 14, ...
 - 2 Find the 15th term of the sequence 7, 12, 17, 22, ...
 - 3 An arithmetic sequence has first term 4 and common difference 6. Find the 20th term.
 - 4 The 5th term of an arithmetic sequence is 23, and the first term is 3. Find the common difference.
 - 5 An arithmetic sequence has first term 7 and common difference 5. Which term equals 82?
 - 6 The 8th term of an arithmetic sequence is 34, and the common difference is 4. Find the first term.
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Geometric sequences

- 7 Find the next term of the sequence 2, 4, 8, 16, ...
 - 8 Find the 6th term of the sequence 3, 6, 12, 24, ...
 - 9 A geometric sequence has first term 5 and common ratio 3. Find the 5th term.
 - 10 The 4th term of a geometric sequence is 54, and the first term is 2. Find the common ratio.
 - 11 A geometric sequence has 2nd term 6 and 3rd term 18. Find the first term and the common ratio.
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Identifying the type of sequence

- 12 Is 5, 10, 20, 40, ... arithmetic, geometric, or neither?
 - 13 Is 6, 10, 14, 18, ... arithmetic, geometric, or neither?
 - 14 Is 1, 4, 9, 16, ... arithmetic, geometric, or neither?
 - 15 Is 100, 90, 81, 72.9, ... arithmetic, geometric, or neither?
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Finding missing terms

- 16 Find the missing term: 3, __, 11, 15 (arithmetic).
- 17 Find the missing term: 4, __, 36 (geometric, all terms positive).
- 18 Insert two arithmetic means between 5 and 20, forming the four-term sequence 5, __, __, 20.
- 19 The sequence x , $2x + 1$, $4x - 1$ is arithmetic. Find x .
- 20 Find the two missing terms of the geometric sequence 2, __, __, 54.
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Sum of an arithmetic series

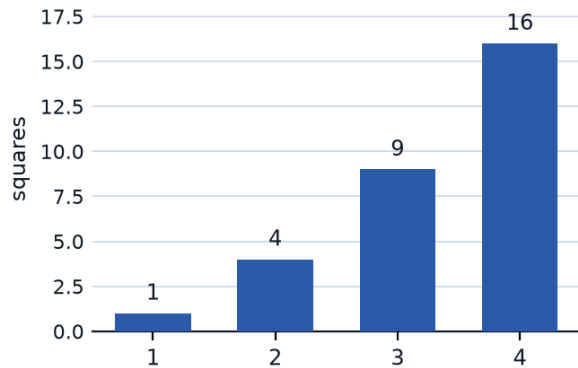
- 21 Find the sum of the first 10 terms of the sequence 2, 5, 8, 11, ...
- 22 Find the sum of the first 20 positive integers.
- 23 Find the sum of the first 12 terms of the sequence 10, 7, 4, 1, ...
- 24 An arithmetic sequence has first term 3 and common difference 4. Find the sum of its first 15 terms.
- 25 Find the sum of the first 25 positive even integers (2, 4, ..., 50).
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Special sequences

- 26 The sequence 1, 4, 9, 16, 25, ... consists of perfect squares. Find the 8th term.
- 27 Find the 6th triangular number, from the sequence 1, 3, 6, 10, 15, ...
- 28 A sequence starts 2, 3, 5, 8, 13, ..., where each term is the sum of the two terms before it. Find the next two terms.
- 29 Find the 7th term of the sequence of powers of 2: 1, 2, 4, 8, 16, ...
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Visual and growing patterns

- 30** The bar chart shows the number of small squares used to build patterns 1 through 4. Predict the number of squares in pattern 6, and state the rule.



- 31** A pattern of dots grows as 3, 7, 11, 15, ... (each stage adds 4 dots). Find the number of dots in stage 10.
- 32** A row of matchstick triangles uses 3, 5, 7, 9, ... matchsticks for 1, 2, 3, 4 triangles. Find the number of matchsticks for 15 triangles in a row.
- 33** A row of matchstick hexagons uses 6, 11, 16, 21, ... matchsticks for 1, 2, 3, 4 hexagons. Find the matchsticks needed for 9 hexagons in a row.

Word problems

- 34** A theater has 20 seats in the first row, and each following row has 3 more seats than the row before it. How many seats are in the 12th row?
- 35** A ball is dropped and after each bounce rises to half its previous height. If it first rises to 80 m after being dropped, how high does it rise after the 4th bounce?
- 36** A savings plan deposits 100 riyals in week 1, then increases the deposit by 20 riyals each following week. Find the total saved after 8 weeks.
- 37** A bacteria culture starts with 50 bacteria and doubles every hour. How many bacteria are there after 6 hours?

More on series and sequences

- 38** Find the sum of the first 8 terms of the geometric sequence 1, 2, 4, 8, ...
- 39** A geometric sequence has first term 4 and common ratio 2. Find the sum of its first 6 terms.

- 40 An arithmetic sequence has 3rd term 11 and 7th term 27. Find the common difference and the first term.
- 41 The n th term of a sequence is $T_n = 3n - 2$. Find the 12th term, and state whether the sequence is arithmetic.
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Pattern reasoning

- 42 A sequence of tile patterns is built so that stage n uses $n^2 + 1$ tiles. How many tiles are needed for stage 7?
- 43 The number of diagonals in a polygon with n sides is $\frac{n(n-3)}{2}$. Find the number of diagonals in an octagon ($n = 8$).
- 44 A sequence follows the rule "double and add 1", starting at 3 as the 1st term. Find the 4th term.
- 45 The number of handshakes when n people each shake hands with every other person once is $\frac{n(n-1)}{2}$. Find the number of handshakes among 10 people.
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MCQ — Arithmetic sequences

- 46 Find the 12th term of the sequence 4, 9, 14, 19, ...
- a. 49
 - b. 54
 - c. 64
 - d. 59
- 47 Find the 18th term of the sequence 6, 13, 20, 27, ...
- a. 120
 - b. 132
 - c. 125
 - d. 118

- 48 An arithmetic sequence has first term 9 and common difference 4. Find the 25th term.
- a. 100
 - b. 105
 - c. 109
 - d. 96
- 49 The 6th term of an arithmetic sequence is 41, and the first term is 6. Find the common difference.
- a. 7
 - b. 8
 - c. 6
 - d. 35
- 50 An arithmetic sequence has first term 10 and common difference 6. Which term equals 100?
- a. 17th
 - b. 90th
 - c. 15th
 - d. 16th
- 51 The 10th term of an arithmetic sequence is 52, and the common difference is 5. Find the first term.
- a. 12
 - b. 2
 - c. 7
 - d. 45

MCQ — Geometric sequences

- 52 Find the next term of the sequence 3, 6, 12, 24, ...
- a. 30
 - b. 27
 - c. 36
 - d. 48
- 53 Find the 7th term of the sequence 2, 6, 18, 54, ...
- a. 486
 - b. 4,374
 - c. 1,458
 - d. 729
- 54 A geometric sequence has first term 4 and common ratio 2. Find the 6th term.
- a. 256
 - b. 32
 - c. 128
 - d. 64
- 55 The 5th term of a geometric sequence is 162, and the first term is 2. Find the common ratio.
- a. 3
 - b. 9
 - c. 4
 - d. 2

- 56 A geometric sequence has 3rd term 20 and 4th term 40. Find the first term and the common ratio.
- a. $a_1 = 5, r = 2$
 - b. $a_1 = 20, r = 2$
 - c. $a_1 = 5, r = 4$
 - d. $a_1 = 10, r = 2$
- 57 A geometric sequence has first term 7 and common ratio 2. Find the 8th term.
- a. 448
 - b. 1,792
 - c. 112
 - d. 896
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MCQ — Identifying the type of sequence

- 58 Is 7, 14, 28, 56, ... arithmetic, geometric, or neither?
- a. Neither
 - b. Arithmetic
 - c. Geometric
 - d. Both arithmetic and geometric
- 59 Is 8, 13, 18, 23, ... arithmetic, geometric, or neither?
- a. Both arithmetic and geometric
 - b. Geometric
 - c. Arithmetic
 - d. Neither

- 60 Is 2, 5, 10, 17, ... arithmetic, geometric, or neither?
- a. Geometric
 - b. Arithmetic
 - c. Both arithmetic and geometric
 - d. Neither
- 61 Is 200, 180, 162, 145.8, ... arithmetic, geometric, or neither?
- a. Both arithmetic and geometric
 - b. Arithmetic
 - c. Geometric
 - d. Neither
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MCQ — Finding missing terms

- 62 Find the missing term: 6, __, 16, 21 (arithmetic).
- a. 9
 - b. 13
 - c. 11
 - d. 10
- 63 Find the missing term: 5, __, 80 (geometric, all terms positive).
- a. 42.5
 - b. 20
 - c. 16
 - d. 40

- 64 Insert two arithmetic means between 8 and 29, forming the four-term sequence 8, __, __, 29.
- a. 15, 22
 - b. 14, 21
 - c. 12.5, 20.5
 - d. 15, 21
- 65 The sequence $2x$, $3x + 4$, $5x + 2$ is arithmetic. Find x .
- a. 4
 - b. 2
 - c. 8
 - d. 6
- 66 Find the two missing terms of the geometric sequence 3, __, __, 81.
- a. 12, 27
 - b. 9, 30
 - c. 27, 9
 - d. 9, 27
- 67 Find the missing term: __, 12, 19, 26 (arithmetic).
- a. 7
 - b. 9
 - c. 4
 - d. 5

MCQ — Sum of an arithmetic series

- 68 Find the sum of the first 12 terms of the sequence 3, 7, 11, 15, ...
- a. 312
 - b. 300
 - c. 264
 - d. 282
- 69 Find the sum of the first 30 positive integers.
- a. 900
 - b. 450
 - c. 435
 - d. 465
- 70 Find the sum of the first 15 terms of the sequence 12, 9, 6, 3, ...
- a. -135
 - b. -153
 - c. -108
 - d. 135
- 71 An arithmetic sequence has first term 5 and common difference 3. Find the sum of its first 20 terms.
- a. 685
 - b. 670
 - c. 640
 - d. 625

72 Find the sum of the first 30 positive even integers (2, 4, ..., 60).

- a. 465
- b. 960
- c. 900
- d. 930

73 Find the sum of the first 10 terms of the sequence 20, 15, 10, 5, ...

- a. -25
- b. -50
- c. 25
- d. 5

MCQ — Sum of a geometric series

74 Find the sum of the first 6 terms of the sequence 2, 4, 8, 16, ...

- a. 128
- b. 124
- c. 126
- d. 62

75 A geometric sequence has first term 5 and common ratio 3. Find the sum of its first 5 terms.

- a. 605.5
- b. 1,210
- c. 363
- d. 605

- 76 Find the sum of the first 4 terms of the sequence 1, 3, 9, 27, ...
- a. 40
 - b. 81
 - c. 27
 - d. 39
- 77 A geometric sequence has first term 6 and common ratio 2. Find the sum of its first 7 terms.
- a. 768
 - b. 762
 - c. 384
 - d. 381

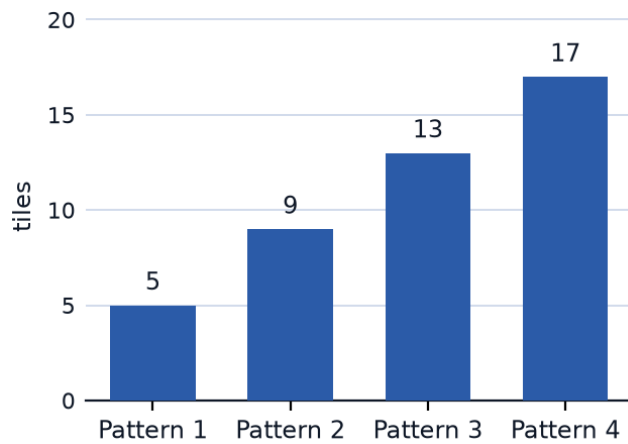
MCQ — Special sequences

- 78 The sequence 1, 4, 9, 16, 25, ... consists of perfect squares. Find the 10th term.
- a. 100
 - b. 121
 - c. 90
 - d. 81
- 79 Find the 7th triangular number, from the sequence 1, 3, 6, 10, 15, 21, ...
- a. 28
 - b. 25
 - c. 36
 - d. 21

- 80 A sequence starts 3, 5, 8, 13, 21, ..., where each term is the sum of the two terms before it. Find the 8th term.
- a. 55
 - b. 89
 - c. 144
 - d. 76
- 81 Find the 8th term of the sequence of powers of 3: 1, 3, 9, 27, 81, ...
- a. 729
 - b. 6,561
 - c. 2,184
 - d. 2,187
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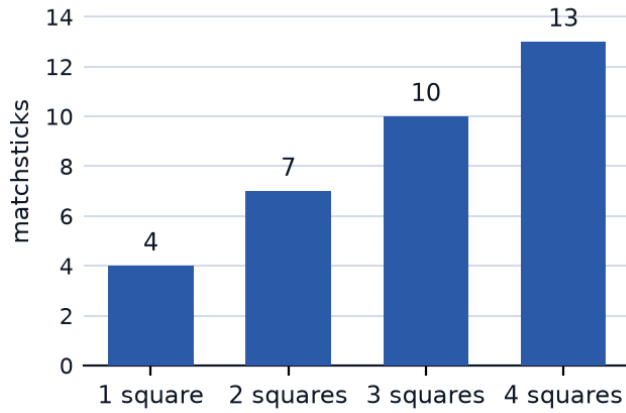
MCQ — Visual and growing patterns

- 82 The bar chart shows the number of tiles used to build patterns 1 through 4. Predict the number of tiles in pattern 8.

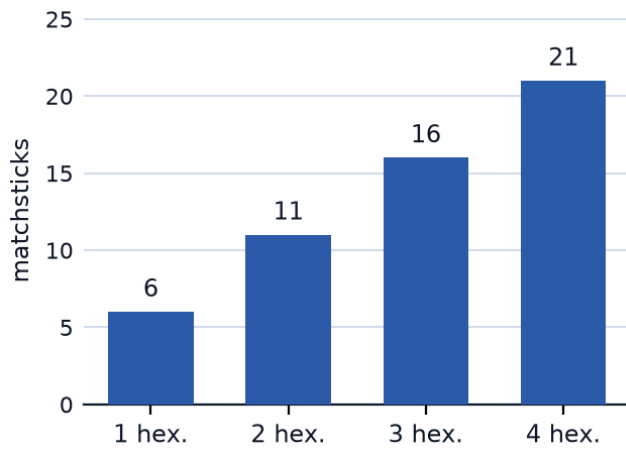


- a. 37
- b. 40
- c. 29
- d. 33
- 83 A pattern of dots grows as 2, 6, 10, 14, ... (each stage adds 4 dots). Find the number of dots in stage 10.
- a. 38
- b. 40
- c. 34
- d. 42

- 84 The bar chart shows the number of matchsticks used to build squares in a row, for 1 through 4 squares. Find the number needed for 12 squares.



- a. 34
- b. 43
- c. 40
- d. 37
- 85 The bar chart shows the number of matchsticks used to build hexagons in a row, for 1 through 4 hexagons. Find the number needed for 9 hexagons.



- a. 41
- b. 51
- c. 50
- d. 46

MCQ — Word problems

- 86** A theater has 18 seats in the first row, and each following row has 4 more seats than the row before. How many seats are in row 15?
- a. 78
 - b. 70
 - c. 56
 - d. 74
- 87** A ball is dropped and after each bounce rises to half its previous height. If it first rises to 120 cm after the first bounce, how high does it rise after the 5th bounce?
- a. 60
 - b. 15
 - c. 3.75
 - d. 7.5
- 88** A savings plan deposits 80 riyals in week 1, then increases the deposit by 15 riyals each following week. Find the deposit in week 10.
- a. 215
 - b. 200
 - c. 230
 - d. 935

- 89 A bacteria culture starts with 40 bacteria and triples every hour. How many bacteria are there after 4 hours?
- a. 1,080
 - b. 3,240
 - c. 9,720
 - d. 480
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MCQ — Pattern reasoning

- 90 A sequence of tile patterns is built so that stage n uses $2n^2 + 1$ tiles. How many tiles are needed for stage 5?
- a. 50
 - b. 26
 - c. 51
 - d. 41
- 91 The number of diagonals in a polygon with n sides is $\frac{n(n-3)}{2}$. Find the number of diagonals in a polygon with 9 sides.
- a. 36
 - b. 27
 - c. 21
 - d. 54

- 92 A sequence follows the rule "triple and subtract 2", starting at 4 as the 1st term. Find the 4th term.
- a. 34
 - b. 28
 - c. 82
 - d. 250
- 93 The number of handshakes when n people each shake hands with every other person once is $\frac{n(n-1)}{2}$. Find the number of handshakes among 12 people.
- a. 66
 - b. 132
 - c. 56
 - d. 78
- 94 The n th term of a sequence is $T_n = 4n - 3$. Find the 15th term, and state whether the sequence is arithmetic or geometric.
- a. 57, geometric
 - b. 53, arithmetic
 - c. 60, arithmetic
 - d. 57, arithmetic
- 95 The n th term of a sequence is $T_n = 5(2)^{n-1}$. Find the 6th term, and state whether the sequence is arithmetic or geometric.
- a. 160, geometric
 - b. 320, geometric
 - c. 160, arithmetic
 - d. 80, geometric

Multiple choice — easy

96 Find the 13th term of the sequence $-14, -21, -28, -35, \dots$

- a. -98
- b. -105
- c. -96
- d. -91

97 Find the 17th term of the sequence $-13, -19, -25, -31, \dots$

- a. -103
- b. -107
- c. -109
- d. -115

98 Find the 20th term of the sequence $1, -3, -7, -11, \dots$

- a. -71
- b. -75
- c. -73
- d. -79

99 Find the 5th term of the sequence $0, 6, 12, 18, \dots$

- a. 26
- b. 24
- c. 30
- d. 18

100 Find the 14th term of the sequence 12, 21, 30, 39, ...

- a. 131
- b. 138
- c. 129
- d. 120

101 Find the 8th term of the sequence 6, 14, 22, 30, ...

- a. 54
- b. 64
- c. 70
- d. 62

102 Find the 19th term of the sequence $-20, -14, -8, -2, \dots$

- a. 82
- b. 90
- c. 88
- d. 94

103 Find the 7th term of the sequence $-5, -1, 3, 7, \dots$

- a. 23
- b. 19
- c. 21
- d. 15

104 Find the 11th term of the sequence $-11, -17, -23, -29, \dots$

- a. -77
- b. -71
- c. -65
- d. -69

105 Find the 6th term of the sequence $-2, 1, 4, 7, \dots$

- a. 10
- b. 13
- c. 16
- d. 15

106 Find the 18th term of the sequence $-5, -13, -21, -29, \dots$

- a. -133
- b. -139
- c. -141
- d. -149

107 Find the 15th term of the sequence $-14, -6, 2, 10, \dots$

- a. 98
- b. 106
- c. 90
- d. 100

108 Find the 10th term of the sequence $2, -7, -16, -25, \dots$

- a. -79
- b. -88
- c. -77
- d. -70

109 Find the 14th term of the sequence $-16, -22, -28, -34, \dots$

- a. -94
- b. -100
- c. -92
- d. -88

110 Find the 15th term of the sequence $4, -4, -12, -20, \dots$

- a. -100
- b. -108
- c. -106
- d. -116

Multiple choice — medium

111 Find the 3rd term of the sequence $-3, 6, -12, 24, \dots$

- a. 12
- b. -10
- c. -14
- d. -12

112 Find the 3rd term of the sequence $-6, 18, -54, 162, \dots$

- a. 54
- b. -54
- c. -52
- d. -56

113 Find the 7th term of the sequence $-1, 3, -9, 27, \dots$

- a. -729
- b. 729
- c. -727
- d. -731

114 Find the 3rd term of the sequence $-2, 4, -8, 16, \dots$

- a. -6
- b. -10
- c. 8
- d. -8

115 Find the 5th term of the sequence $-8, -24, -72, -216, \dots$

- a. -650
- b. -648
- c. 648
- d. -646

116 Find the 7th term of the sequence $-5, 10, -20, 40, \dots$

- a. -320
- b. 320
- c. -322
- d. -318

117 Find the 5th term of the sequence $3, -9, 27, -81, \dots$

- a. -243
- b. 245
- c. 241
- d. 243

118 Find the 6th term of the sequence $6, 12, 24, 48, \dots$

- a. 194
- b. 190
- c. 192
- d. -192

119 Find the 5th term of the sequence $-1, 3, -9, 27, \dots$

- a. -79
- b. -83
- c. 81
- d. -81

120 Find the 6th term of the sequence $-1, 3, -9, 27, \dots$

- a. 241
- b. 243
- c. -243
- d. 245

121 Find the 5th term of the sequence $-5, -15, -45, -135, \dots$

- a. -405
- b. -403
- c. 405
- d. -407

122 Find the 3rd term of the sequence $-5, 15, -45, 135, \dots$

- a. -43
- b. 45
- c. -47
- d. -45

123 Find the 3rd term of the sequence $7, 21, 63, 189, \dots$

- a. 61
- b. 65
- c. 63
- d. -63

124 Find the 3rd term of the sequence $-1, -3, -9, -27, \dots$

- a. 9
- b. -9
- c. -7
- d. -11

125 Find the 7th term of the sequence $-6, -18, -54, -162, \dots$

- a. -4372
- b. -4374
- c. 4374
- d. -4376

Multiple choice — hard

126 Find the sum of the first 14 terms of the sequence $-3, -11, -19, -27, \dots$

- a. -766
- b. -774
- c. -778
- d. -770

127 Find the sum of the first 18 terms of the sequence $-3, -9, -15, -21, \dots$

- a. -968
- b. -976
- c. -978
- d. -972

128 Find the sum of the first 11 terms of the sequence $-1, -4, -7, -10, \dots$

- a. -179
- b. -176
- c. -180
- d. -172

129 Find the sum of the first 19 terms of the sequence $12, 9, 6, 3, \dots$

- a. -288
- b. -289
- c. -285
- d. -281

130 Find the sum of the first 20 terms of the sequence $12, 13, 14, 15, \dots$

- a. 426
- b. 431
- c. 430
- d. 434

131 Find the sum of the first 6 terms of the sequence $-4, -8, -16, -32, \dots$

- a. -255
- b. -249
- c. -256
- d. -252

132 Find the sum of the first 6 terms of the sequence $-7, -21, -63, -189, \dots$

- a. -2545
- b. -2551
- c. -2548
- d. -2555

133 Find the sum of the first 6 terms of the sequence $5, -10, 20, -40, \dots$

- a. -108
- b. -102
- c. -100
- d. -105

134 Find the sum of the first 6 terms of the sequence $-3, 9, -27, 81, \dots$

- a. 543
- b. 549
- c. 546
- d. 552

135 Find the sum of the first 5 terms of the sequence $9, -27, 81, -243, \dots$

- a. 546
- b. 558
- c. 552
- d. 549

136 Find the missing term: -1 , $_\$, -15 , -22 (arithmetic).

- a. -1
- b. -8
- c. -15
- d. -6

137 Find the missing term: 4 , $_\$, 22 , 31 (arithmetic).

- a. 4
- b. 15
- c. 13
- d. 22

138 Find the missing term: 3 , $_\$, 27 (geometric, all terms positive).

- a. 6
- b. 30
- c. 12
- d. 9

139 Find the missing term: 1 , $_\$, 9 (geometric, all terms positive).

- a. 3
- b. 6
- c. 10
- d. 1

140 Find the missing term: 5, __, 20 (geometric, all terms positive).

a. 10

b. 25

c. 12

d. 8